

Charlie Marx

CURRICULUM VITAE - DECEMBER 2019

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EDUCATION	Haverford College B.S. in Computer Science (Major GPA: 3.97/4.0) B.S. in Mathematics (Major GPA: 4.0/4.0) Minor in Statistics <i>Relevant Coursework:</i> Machine Learning · Linear Algebra · Algorithms Advanced Topics in Probability · Abstract Algebra · Real Analysis Programming Languages · Multivariable Calculus · Theory of Computation	2016 - MAY 2020 (EXPECTED) GPA: 3.94/4.0
AWARDS & HONORS	Beckman Scholars Award National award for excellence in research including \$26,000 of research and travel funding. Brian Kovaric Award Finalist Faculty-nominated scholarship awarded to one outstanding first-year student. Farmers Insurance Standout Student of the Year Regional award for academic and community engagement awarded to one student each year.	2018 2017 2016
PUBLICATIONS	MACHINE LEARNING METHODS 1. Disentangling Influence: Using Disentangled Representations to Audit Model Predictions Charles T. Marx , Richard L. Phillips, Sorelle A. Friedler, Carlos Scheidegger, Suresh Venkatasubramanian <i>NeurIPS – Conference on Neural Information Processing Systems</i> , 2019 2. Predictive Multiplicity in Classification Charles T. Marx , Flavio du Pin Calmon, Berk Ustun <i>NeurIPS Workshop on Human-Centered Machine Learning</i> , 2019 Accepted for Oral Presentation MACHINE LEARNING APPLICATIONS 3. Examining Inputs of Biogenic and Oil-Derived Hydrocarbons in Surface Waters Following the Deepwater Horizon Oil Spill Helen White, Charles T. Marx , David Valentine, Charles Sharpless, Christoph Aeppli, Kelsey Gosselin, Veronika Kivenson, Rachel Liu, Robert Nelson, Sean Sylva, Christopher Reddy <i>ACS Earth and Space Chemistry</i> , 2019 4. Rapid Identification of Deepwater Horizon Oil Residues Using X-Ray Fluorescence Anna Michel, Alexandra E. Morrison, Charles T. Marx , Helen White <i>ACS Environmental Science & Technology Letters</i> , 2018	
RESEARCH EXPERIENCE	Research Assistant in Machine Learning <i>Harvard University</i> Studied predictive multiplicity: the ability for datasets to admit multiple equally accurate models with conflicting predictions. Developed integer programming tools and software to measure the severity of multiplicity. Compiled findings into first-author conference and workshop paper. <i>Advisors:</i> Flavio Calmon and Berk Ustun Research Assistant in Machine Learning <i>Haverford College</i> Designed new methods to quantify indirect feature influence in machine learning models using disentangled representations and adversarial learning. <i>Advisor:</i> Sorelle Friedler Research Assistant in Chemistry <i>Haverford College</i> Designed machine learning tools to optimize model performance, robustness, and interpretability for on-site oil identification. <i>Advisor:</i> Helen White	SUMMER 2019 2017 – PRESENT 2017 – 2018

	Summer Research Intern <i>Prince William Sound Science Center</i> SUMMER 2018 Automated salmon and herring tracking in Alaska using machine learning and sonar data. <i>Advisor:</i> Scott Pegau
	Research Assistant in Computer Vision <i>Haverford College</i> 2017 Analyzed 100,000 handwritten texts using recurrent and convolutional neural networks. <i>Advisor:</i> Andrew Janco
ACADEMIC ACTIVITIES	Teaching Assistant in Machine Learning 2019 – PRESENT Held weekly office hours and graded for an upper-level machine learning course. Peer Tutor 2018 – 2019 Tutored students in statistics, linear algebra, and machine learning. Computer Science Faculty Search Committee 2018 – 2019 Student representative on the computer science tenure-track faculty search committee.
INVITED & CONTRIBUTED TALKS	Simons Institute for the Theory of Computing <i>Berkeley, CA</i> 2019 Disentangling Influence (45-minute talk) INFORMS Annual Meeting <i>Seattle, WA</i> 2019 Predictive Multiplicity in Classification (20-minute talk) FAT* Conference <i>New York, NY</i> 2018 Auditing Black Box Models (60-minute tutorial) <i>with Carlos Scheidegger and Suresh Venkatasubramanian</i> GOMOSSES Conference <i>New Orleans, LA</i> 2018 Examining Inputs of Biogenic and Oil-Derived Hydrocarbons (15-minute talk)
POSTER PRESENTATIONS	NeurIPS Conference <i>Vancouver, BC</i> 2019 Disentangling Influence: Using Disentangled Representations to Audit Model Predictions NeurIPS Human-Centered Machine Learning Workshop <i>Vancouver, BC</i> 2019 On the Multiplicity of Predictions in Machine Learning Beckman National Symposium <i>Irvine, CA</i> 2019 Indirect Influence in Machine Learning Models Beckman Regional Symposium <i>New York, NY</i> 2018 Mitigating Social Discrimination in Deep Learning KINSC Undergraduate Research Symposium <i>Haverford, PA</i> 2017 Examining Biogenic Hydrocarbons After the Deepwater Horizon Oil Spill
FIELDWORK	Alaska Prince William Sound Research Charter 2018 Gulf of Mexico Shoreline Research Tour 2017
EXTRA- CURRICULAR ACTIVITIES	Captain of Haverford College Men’s Club Soccer 2018 – 2019 Leader of Haverford College Outdoors Club 2017 – 2018 President of Furniture for Newly Housed Families Organization 2015 – 2017
SKILLS & INTERESTS	Languages: <i>Fluent:</i> Python, R, LaTeX · <i>Familiar:</i> SQL, C, Unix Tools: Keras/Tensorflow, PyTorch, Pandas, CPLEX Interests: Backpacking, Gardening, Soccer, Folk Music, Cooking